



Guided Research Activity
Kitchen Curiosity

Overview	In this lesson, students investigate the properties of everyday household substances by observing their physical qualities and testing them with a natural pH indicator. After predicting whether each liquid is acidic, neutral, or basic, they use red cabbage water or litmus solution to test their hypotheses and record the color changes. Students compare their predicted results with the experimental outcomes to learn how acids and bases are classified on the pH scale. This activity helps students connect their prior experiences with substances like vinegar, soap, and lemon juice to the scientific concept of pH and chemical reactivity.
Objective/ Goal	Students will know: ● That substances can be classified as acids, bases, or neutral based on their pH. ● That the pH scale ranges from 0–14, with values below 7 being acidic, above 7 being basic, and 7 being neutral. ● That indicators such as red cabbage solution or litmus change color in the presence of acids or bases. ● That everyday household items provide real-life examples of acids and bases. Students will do: ● Observe and describe the physical properties of common household substances. ● Predict whether each substance is acidic, neutral, or basic based on prior knowledge and experience. ● Test each substance with a natural indicator solution and record the color changes. ● Compare hypotheses with results to determine whether predictions were supported. ● Discuss how their findings connect to real-world examples of acids and bases.
Key Words	● Acid ● Base



	• pH scale • Neutral • Litmus
US Curriculum Standards:	5th grade: • 6th grade: • 7th grade: • 8th grade: •
Difficulty Level	Easy
Materials	• Kitchen Curiosity Lab Sheet • water • A mix of kitchen/household substances ○ Vinegar/lemon juice ○ Soap ○ Potassium/Sodium Hydroxide ○ Rubbing alcohol/hand sanitizer • Test tubes for each substance • Red Cabbage juice/litmus solution • Transfer pipettes. • Colored pencils
Estimated Time	45 min

Investigation Procedure		
	Description/Key Points	Points
Observation	Liquid Observations	

	Gather all of the kitchen/household substances for testing. Be sure to have a mix of acids and bases including: vinegar, soap, sodium or potassium hydroxide (equivalent of Drano), water (control) and alcohol (or hand sanitizer). Students “observe” each object noting some of their qualities including color, smell, taste (based on experience not actual tasting), and texture. If using potassium/sodium hydroxide DO NOT TOUCH Instruct students to write the name of each substance on their lab sheet as they make their observations.	
Idea & Questions	Discuss the differences between acids and bases. Ask students for their experiences with each. Examples: - What does tasting a lemon feel like? - What about soap if you’ve gotten soap in your mouth? Show the red cabbage water or litmus solution and explain what is going to happen to the color when different substrates are added. Have students color in the ‘Before Color’ column on the data sheet (all will be the same color). Explain that every substance is classified with a pH value between 1-14. Numbers less than 7 are acidic, numbers greater are basic. Explain that 7 is neutral. Have students drop 5ml of water into a test tube of litmus solution. Does the color change?	
Hypothesis	Based on your observations, in the ‘Hypothesis Color’ column, color in the color you hypothesize each test substance will turn once mixed with the litmus solution. Please use the guide at the top of the datasheet to help you determine this color.	

	Example: <i>"I hypothesize that the deodorant will be a strong base because it feels soapy and slippery"</i>	
Experiment/Test	1. Set up enough test tubes for each test liquid. 2. Pipette between 5-10ml of litmus solution into the test tube using a transfer pipette. 3. Pipette a small amount (5ml or less) of the substance you are testing into the tube 4. Gently swirl the solution for at least 10 seconds being careful not to spill the solution. a. If you add a pellet to the tube you will need to mix for longer until the pellet is fully dissolved. 5. Note the color change (if any) and in the 'After Color' column fill in the color of the final solution in your tube 6. Place the tube in the rack and note the results of the other substances.	
Results & Analysis	Review the 'Hypothesis Color' column and note if your hypothesis for each substance was supported or not supported by the experiment data ('After Color' column). In the 'Hypothesis Supported?' column note whether your data supported the hypothesis. ● If the actual versus the hypothesized colors were different then the hypothesis was not supported and you should write 'NO' in the column. ● If the colors are the same then your hypothesis was supported by the experiment data and you should write 'YES' in the column. Discuss: ● What are other examples of acids and bases that are in your everyday life? Make a few predictions based on these experimental results ○ Test in time allows	



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| | <ul style="list-style-type: none">• What effect could acids and bases have on water if they get into streams or lakes? | |
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